

What a Language Model Does with a Solver Certificate: An Answer-Evidence Decomposition Across Twelve Models

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Abstract

Solver-, tool- and retrieval-augmented systems report large accuracy gains, routinely read as evidence that the model used the supplied material. Two mechanisms give the same number: reading a supplied answer, and reasoning over supplied state. The standard control, a same-shape irrelevant record, separates neither. We build one that does, holding every surface property we could enumerate and audit fixed while destroying validity alone, across four sealed experiments and two robustness studies: 34,584 scored responses, re-derivable from released receipts.

The decomposition is the contribution; what it returns is not. The validity share is 98.3% on one checkpoint and 40.4–51.4% on another, so a single-model mechanism result should not be generalised, ours included. Two negative results hold on every checkpoint: reversing a rule and the premise it fires on costs accuracy, and all three follow a fabricated final rule to the wrong answer on 189, 186 and 192 of 192 items. We find no positive evidence that any checkpoint verifies validity (the strongest abstains on 0 of 192 broken certificates), but report that the abstention test’s direction depends on which reference arm it uses.

We report the design’s limits too: the 2×2 is ceiling-limited, viability is a property of (model $\times$ arm $\times$ runtime) not of a model, and two prespecified criteria of our own were invalidated by our own data.

1 Introduction

Pairing a language model with an external solver, tool or retrieval index reliably raises accuracy on tasks the model cannot do alone. Systems that translate a problem into a formal representation, dispatch it to a symbolic engine and return the verdict through the model report large end-to-end gains [Pan et al., 2023, Olausson et al., 2023, Ye et al., 2023, Gao et al., 2022, Schick et al., 2023, Yang et al., 2023, Lewis et al., 2020]. Those gains are routinely read as evidence that the model *used* the supplied material.

That reading is underdetermined. Two mechanisms produce the same accuracy number. Under *answer reading*,

the supplied record contains the conclusion, or something that co-occurs with it, and the model copies out the implied response. Under *state use*, the model works with the supplied premises and completes the remaining inference itself. The first says nothing about what the model does with reasoning; the second says something specific. The standard control is a same-shape but irrelevant record, which rules out proof length, generic validity and formatting. It does not rule out answer reading, because a *relevant* certificate stays answer-diagnostic even when the response token is withheld: a prefix stopping one step short still names the query’s subject and predicate in adjacent lines, so a model that only checks surface co-occurrence reproduces every number a reasoning model would.

Separating the two requires a control that holds surface form fixed and destroys validity alone. This paper builds one and applies it across four sealed experiments on frozen small models, scoring by direct next-token likelihood over verified single-token candidates so that no free text is parsed, with two robustness studies on line order and on the inference runtime.

The method is the contribution; what it returns is model-specific. Breaking validity alone removes almost the whole state effect on one checkpoint (a validity share of 98.3%) and 40.4% on another, which answers 58 of 96 broken certificates correctly where the first manages 1. We offer the decomposition and decline to generalise the value it returns, ours included. Two results do hold everywhere: accuracy is degraded when the final rule precedes the premise it fires on, and none of the three checkpoints resists a corrupted apparatus, answering incorrectly on 189, 186 and 192 of 192 items given a certificate whose fabricated final rule establishes the negation. Depth is checkpoint-specific, and whether the model detects invalidity our abstention test cannot settle in either direction.

Two measurement points shape how this may be reported. Accuracy alone is not interpretable: one arm scored 50.0% on a balanced panel at $d' = 0.00$, so sensitivity and criterion accompany it throughout [Macmillan and Creelman, 1991, Stanislaw and Todorov, 1999]. And the claim boundary is strict: on the Qwen checkpoints every arm without a near-complete chain returns $d' = 0$, so those

models do not reason unaided and high accuracy with full solver material describes a faithful transcriber. Gemma-3-4b is not in that position (22/96 unaided), which is why its share is reported against two baselines. System-level effects are never upgraded into claims about the model’s own reasoning.

What is new is the instrument: an answer-evidence decomposition that separates solver-supplied answers from solver-supplied state, a surface-matched invalidity control certified per item, and a forward-closure certifier for underdetermined items. We apply it to twelve models [Qwen Team et al., 2024, Grattafiori et al., 2024, Gemma Team et al., 2025, Microsoft et al., 2025, OLMo Team et al., 2024, Allal et al., 2025].

2 Related Work

2.1 Faithfulness, and the solver-augmented pipeline

Chain-of-thought made intermediate steps part of the output [Wei et al., 2022], and a literature then asked whether the answer rests on them: explanations omit the biasing features that demonstrably drive the prediction [Turpin et al., 2023]; truncating a trace and injecting mistakes into it reveals wide variation in conditioning, with injected mistakes often ignored [Lanham et al., 2023, Arcuschin et al., 2025]; models trained for long traces verbalise answer-determining hints only a minority of the time [Chen et al., 2025]; and one response is architectural, making the trace a program a solver executes [Lyu et al., 2023]. Neurosymbolic pipelines put the formal step outside the model altogether: translating into a symbolic language and repairing from solver errors [Pan et al., 2023], parsing into first-order logic for a prover [Olausson et al., 2023], dispatching a declarative specification [Ye et al., 2023], emitting a program for an interpreter [Gao et al., 2022]. The pattern extends to self-supervised API use [Schick et al., 2023], premise selection [Yang et al., 2023] and retrieval [Lewis et al., 2020].

Both leave our question open. The first intervenes on *model-generated* traces, asking whether the model’s own account is honest; ours is *externally supplied, formally verified* state, where the question inverts, and where ignoring a mistake in one’s own trace is unfaithful, whereas following a corrupted external certificate is unsafe. The second reports end-to-end accuracy, which cannot separate the solver’s *answer* from its *state*, and assumes the solver correct, though the model side is not neutral to context [Shi et al., 2023, Xie et al., 2023]. We decompose the two channels and add a surface-matched invalid certificate and a valid-looking one that points the wrong way.

2.2 Benchmarks, and the limits of accuracy

The task family descends from rule-based deductive datasets in natural language: RuleTaker [Clark et al., 2020], ProofWriter, which added open-world semantics with an explicit unknown class [Tafjord et al., 2020], PrOntoQA [Saparov and He, 2022] and FOLIO [Han et al., 2022]. Order effects are large in deductive prompting [Chen et al., 2024]; these sets predate the models evaluated on them, so contamination cannot be excluded [Deng et al., 2024, Golchin and Surdeanu, 2023]; and counterfactual variants show how much apparent competence rests on familiarity [Wu et al., 2023]. None supplies the matched adversarial certificate variants an attribution question needs, so we generate items after model release with per-item nonce vocabularies and build those certificates ourselves.

Prompted classification is biased rather than merely noisy: few-shot accuracy swings from near chance to near state of the art with format and example order, because the model carries a prior toward particular answer strings [Zhao et al., 2021]; option order alone moves multiple-choice accuracy [Pezeshkpour and Hruschka, 2023, Zheng et al., 2023]; likelihood scoring adds a confound, since strings compete for probability mass independently of correctness [Holtzman et al., 2021]; and abstention is its own capability [Kadavath et al., 2022, Yin et al., 2023]. Solver-augmented evaluation has not absorbed this: on a balanced panel a degenerate single-label responder is indistinguishable from a chance-level competent one. Signal detection theory separates them [Macmillan and Creelman, 1991, Stanislaw and Todorov, 1999], so we report d' and criterion c for every arm, with the exact McNemar test [McNemar, 1947] and bootstrap intervals [Efron, 1987]. Two concrete cases here motivate the practice rather than illustrate it: an arm scoring 50.0% on a balanced panel has $d' = 0.00$, and a prespecified criterion on a single class’s recall was passed by models answering one label almost everywhere. One is a criterion of our own, replaced post hoc and labelled as such.

3 Method

3.1 The answer-evidence decomposition

A solver certificate carries two things at once: the *answer* to the query, and the *state* from which that answer follows. End-to-end accuracy conflates them. We separate them by arranging arms as a 2×2 of answer evidence by proof state, with a no-material baseline outside the square.

Let `IRRELEVANT` supply a valid derivation for a different entity (no answer, no relevant state), `CONCLUSION_ONLY` the derived terminal literal alone (answer, no state), `PROOF_PREFIX` the chain with that literal withheld

(state, no answer), and FULL both. An *edge* of this square moves one factor; the *diagonal* moves two, so a null on it attributes to neither. This is identification, not statistics: the prespecified primary contrast of the experiment we re-analyse, `PROOF_PREFIX` minus `CONCLUSION_ONLY`, is that diagonal. The four edges instead give a state main effect (the mean of the two edges adding state) and an answer main effect defined symmetrically.

3.2 Surface-matched invalidity

A prefix that stops one step short is not a clean state manipulation. It still names the query’s subject and predicate in adjacent lines, so surface co-occurrence remains available. To isolate validity we construct `BROKEN_CHAIN`: one intermediate link is cut so the displayed lines no longer reach the query, while every surface property we enumerate below is held fixed.

Concretely, against the matched `TRUNCATE_1` arm we preserve the query-entity occurrence count (audited per item, not approximately), the presence of the query predicate in the final rule, the line count and the exact fact/rule/literal type sequence. Per-item character lengths are recorded but not constrained. Every *rule* displayed is a real rule of the theory; exactly one derived literal is false. All 192 items are certified by forward-closure saturation (section 3.3) to leave the query *undetermined* under the displayed lines. The decomposition follows:

$$\text{total state} = \text{TRUNCATE_1} - \text{IRRELEVANT}, \quad (1)$$

$$\text{surface} = \text{BROKEN_CHAIN} - \text{IRRELEVANT}, \quad (2)$$

$$\text{validity} = \text{TRUNCATE_1} - \text{BROKEN_CHAIN}. \quad (3)$$

Four companion arms separate the remaining confounds. `SAME_ENTITY_IRRELEVANT` gives a valid derivation about the *query* entity toward an unrelated predicate. It names the query subject four times where `IRRELEVANT` names it zero, and neither contains the query predicate, so the pair isolates entity repetition. `SHUFFLED` presents `TRUNCATE_1`’s lines in seeded random order; `TRUNCATE_2` and `TRUNCATE_3` withhold two and three steps, giving a depth ladder; and `MISLEADING` supplies a valid-looking chain whose single fabricated final rule establishes the query’s negation, leaving the ground-truth answer unchanged so that following the supplied state means answering incorrectly.

3.3 Certifying underdetermination

Showing that a query *is* derivable is easy: exhibit the proof. Showing it is *not* requires saturating the theory’s forward closure. Our certifier operates under open-world semantics with explicit negation: a negative literal is derivable

only if something derives it, never by failure to prove the positive. Antecedents are ordinary literals, with no negation as failure, which makes naive forward chaining sound and complete here.

Two safeguards matter for generated theories: a round cap with a saturated flag, so a malformed theory fails loudly rather than spinning, and rejection of any theory deriving both an atom and its negation. An item is *undetermined* when neither the query nor its negation appears in the closure.

Validation has three layers, because one would not be worth much. Against the open-world splits of a public corpus [Tafjord et al., 2020] at depths 0–3 and 5, 23,240 of 23,240 questions agree, including 10,440 undetermined. But those labels also come from forward chaining over the same fragment, so this largely measures one chainer against another and covers none of the nonce theories. Nine adversarial theories therefore probe cycles, chains past the round cap, derived negation, absent symbols and entity-specific rules; all nine pass and the cases are released. A *differential* test against an independently written reference agrees on 192/192 items, which catches implementation error but not a misconception about the semantics, since it encodes the same ones. No item enters a sealed panel without a certification record.

3.4 Measurement: sensitivity, not accuracy

Accuracy on a balanced panel is uninterpretable when the responder is biased. One arm below scores 50.0% with an identical 9/96 “Yes” rate in *both* classes: $d' = 0.00$, no discriminative signal at all, at or below the trivial always-one-label strategy. We therefore report sensitivity d' and criterion c alongside accuracy for every arm [Macmillan and Creelman, 1991, Stanislaw and Todorov, 1999], using the loglinear correction for extreme rates. For the three-class panel the headline scalar is balanced accuracy, the mean of per-class recalls: chance is 33.3%, and a single-label responder scores exactly chance.

Paired contrasts report the complete 2×2 table, the exact two-sided McNemar test [McNemar, 1947] as the decision criterion, and a seeded 20,000-resample paired bootstrap interval (bias-corrected and accelerated for paired differences [Efron, 1987], percentile for the share ratio, which is a ratio of differences), with Holm adjustment within declared families.

4 Experimental Setup

4.1 Task, panels and scoring

Every experiment and study here uses rule-chaining queries over small theories of facts and universally quantified rules. Table 5 lists the panels. The sealed panels are procedurally generated after model release with per-item nonce vocabularies, are mutually disjoint, and fix derivation depth at four by construction, recorded per item. The public corpus is used *only* to validate the certifier; no item from it is scored by any model here.

Experiment 4 generates no panel: it rescores Experiment 2’s items with a third candidate available. On Experiment 3’s 64 undetermined items there is no terminal literal to supply, so `CONCLUSION_ONLY` states that the query is not derivable and `PROOF_PREFIX` is truncated at a random point; those two cells instantiate the 2×2 less exactly than the other thirds. Theories, certificates and queries are byte-identical; only the instruction line and candidate set change, and the change is recorded in the seal. Its `NONE` arm withholds the record under that same instruction, fixing the abstention floor.

Two robustness studies reuse these panels. Section 5.3 re-scores Experiment 2’s items under eight permutations of the certificate’s lines each, recording where the final rule and its premise land. Section 5.8 re-scores Experiment 3’s panel through a second inference stack, transformers on CUDA rather than `mlx-lm`, and there extends the sweep with Llama-3.1-8B and Qwen2.5-14B.

Responses are scored by direct next-token likelihood over verified single-token candidates: Yes/No for Experiments 1 and 2, Yes/No/Unknown for Experiments 3 and 4. The model generates no free text, so nothing is parsed and no verdict is inferred from prose. Every prompt is instruction, theory, solver record and query, with the record line omitted entirely in `NONE`; the theory is visible in all arms. In Experiment 1 the arms differ in the record’s header as well as its content, one of which names the truncation, a confound on that experiment’s prespecified primary contrast, removed in Experiments 2–4.

4.2 Models

Experiments 1 and 2 use the same frozen 3B instruction-tuned checkpoint at 4-bit precision [Qwen Team et al., 2024], pinned to byte-identical weights, so the anchors of the first transfer to the second without a weights caveat. Experiment 3 scores ten instruction-tuned models from 0.5B to 7.6B [Qwen Team et al., 2024, Grattafiori et al., 2024, Gemma Team et al., 2025, Microsoft et al., 2025, OLMo Team et al., 2024, Allal et al., 2025], all loaded through `mlx-lm` from the pinned mirrors in `models.toml`. The

runner did not record a dtype, so we do not claim uniform precision across the ten. An eligibility screen confirmed that all ten load, complete a forward pass, and tokenise all three candidates as single distinct tokens.

4.3 Sealed protocol

Scoring takes the highest-likelihood candidate token; on an exact tie the first candidate in the declared order wins, which across Experiment 2’s 5,760 prospective responses occurs twice. Panels are hash-stamped before scoring; Experiments 2 and 3 refuse a panel whose hash does not match, while the others record the hash without enforcing it. The answer authority is opened after all receipts are written, but by convention in the analysis order rather than by a gate in the code, and we state it as a convention.

Experiment 3 makes a deliberate protocol change: *no gate blocks the run*. Where earlier practice excluded a model that failed a delivery screen, delivery is itself the capability under test here, so a model that cannot emit a verdict reliably is a finding about substrate viability rather than a reason to drop it. Tooling failures (a model that will not load, or whose tokeniser cannot represent the candidates) remain exclusions and are not findings; the screen found none.

5 Results

5.1 Reading the factorial as a 2×2

Figure 1 arranges Experiment 1’s five arms by section 3.1. Its prespecified primary contrast, `PROOF_PREFIX` – `CONCLUSION_ONLY` = -4.7pp (exact McNemar $p = 0.078$), was reported as a null. It is the diagonal.

The four edges are reported in table 1. **This reading is post hoc:** a different analysis of sealed data, not a re-run. Every value is re-derived from the 960 raw receipts by a script that asserts it reproduces all five published per-arm counts before reporting. The state main effect is $+22.4\text{pp}$ and the answer main effect $+27.1\text{pp}$.

The **interaction is -29.2pp** , larger than either main effect. The design is also **ceiling-limited**: `FULL` reaches 191/192 (99.5%), only 13.0pp above `PROOF_PREFIX`’s 86.98%. Both “other factor present” edges are therefore compressed, and the averaged main effects deflated with them.

The answer literal appears to retain 30% of its value once state is present (12.50/41.67), but with `FULL` at ceiling that ratio measures the headroom left rather than redundancy, so we draw no conclusion from it. Separating the channels needs `FULL` away from ceiling.

Table 1: The four one-factor edges of the 2×2 , and the prespecified diagonal. Post hoc reanalysis of sealed data.

Contrast	Effect	Exact McNemar p	Reported originally
Add answer, no state present	+41.67pp	1.7×10^{-24}	no
Add state, no answer present	+36.98pp	1.5×10^{-19}	secondary
Add answer, state present	+12.50pp	1.2×10^{-7}	no
Add state, answer present	+7.81pp	6.1×10^{-5}	secondary
<i>Diagonal (prespecified primary)</i>	-4.69pp	0.078	primary

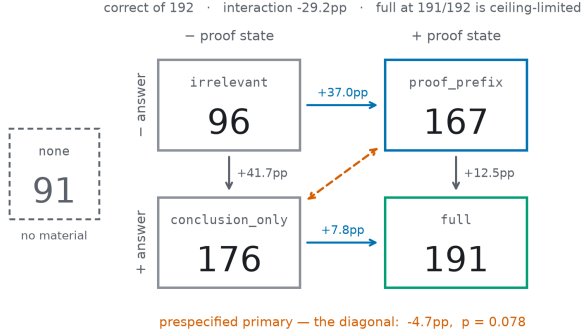


Figure 1: The five arms form a 2×2 of answer evidence by proof state, with the no-material baseline outside it. Each cell gives correct items of 192. Arrows mark the four one-factor edges; the dashed diagonal is the prespecified primary, which moves both factors at once and therefore attributes its null to neither.

Table 2: Signal detection for Experiment 1; hits and false alarms out of 96 each. Apparent chance and zero sensitivity are not the same thing.

Arm	Hits	FA	d'	c
NONE	9	14	-0.25	+1.17
IRRELEVANT	9	9	0.00	+1.29
CONCLUSION_ONLY	80	0	+3.52	+0.81
PROOF_PREFIX	71	0	+3.20	+0.97
FULL	95	0	+4.72	+0.20

Splitting by class overturns the baseline description. The model answers No to 88% of items, and table 2 shows IRRELEVANT returning an identical 9/96 “Yes” rate in both classes at $d' = 0.00$, below a trivial always-No strategy’s 96/192. The contradicted class contributed 0 of the 21 discordant pairs, so the effective sample was 96.

5.2 Validity versus surface overlap

Experiment 2 holds surface form fixed and destroys validity alone. The primary contrast is TRUNCATE_1 – BROKEN_CHAIN = +60.4pp: exact McNemar $p = 6.9 \times 10^{-18}$, BCa [+49.0, +68.8], from a paired table of 1 both correct, 58 TRUNCATE_1-only, 0 BROKEN_CHAIN-only and

37 neither, on the entailed subset ($n = 96$) as declared before the run since the contradicted class contributes no discordance. Sensitivity moves with it (d' drops 2.446), so this is a change in discrimination, not criterion.

The decomposition is therefore a total state effect of +61.5pp comprising +1.0pp of surface overlap ($p = 1$, 95% percentile [+0.0, +3.1]pp) and +60.4pp of validity: a **validity share of 98.3%, 95% percentile** [94.4, 100.0], bootstrapping the whole ratio rather than its numerator. Table 3 and fig. 2 give the full arm set, and section 5.4 shows the share is specific to this checkpoint.

Three supporting contrasts sharpen the reading. Entity repetition explains nothing: SAME_ENTITY_IRRELEVANT – IRRELEVANT = +0.0pp ($p = 1$), despite the former naming the query subject four times and the latter not at all. Order matters *on this checkpoint*: TRUNCATE_1 – SHUFFLED = +46.9pp ($p = 6.8 \times 10^{-13}$, Holm 1.4×10^{-12}), so it is not reading a bag of statements, though section 5.4 shows this too is checkpoint-specific, and section 5.3 shows *order* is the wrong word for it. And depth is a cliff, not a slope: TRUNCATE_2 – TRUNCATE_1 = -61.5pp ($p = 3.5 \times 10^{-18}$, Holm 1.0×10^{-17}), with TRUNCATE_2 scoring identically to supplying no certificate at all. Holm adjustment is within the family declared in section 3.4; table 1 carries the adjusted values for the four post hoc edges. Five arms are indistinguishable at 96/192, 0/96 entailed, $d' = 0.000$, $c = 2.565$: behaviour is binary: the chain reaches one step from the answer, or the model is blind to it.

5.3 Not adjacency, but direction

The gap-versus-direction split here is post hoc. SHUFFLED is one seeded permutation per item, and scrambling moves three things at once: the gap between the final rule and the premise it fires on, the order of that pair, and everything else. We re-scored the same items under eight permutations each on all three checkpoints, recording where those two lines landed. An identity arm reproduces TRUNCATE_1’s bytes and returns its receipts 192/192 throughout. Over eight permutations the effect spans [+41.7, +50.0], [+60.4, +63.5] and [+6.2, +13.5]pp on the three checkpoints, each range containing its published single draw (+46.9, +62.5, +10.4pp).

Table 3: Experiment 2: ten matched arms scored on three checkpoints, entailed subset (correct of 96, with signed d'). Only the solver record differs between arms. The share rows give the validity component as a fraction of the total state effect, against the IRRELEVANT control and against the unaided NONE arm; the two coincide wherever unaided competence is zero.

Arm	Qwen2.5-3B 4-bit		Qwen2.5-3B bf16		Gemma-3-4B	
	/96	d'	/96	d'	/96	d'
NONE	0	0.00	0	0.00	22	1.83
IRRELEVANT	0	0.00	0	0.00	2	0.62
SAME_ENTITY_IRRELEVANT	0	0.00	0	0.00	3	0.77
TRUNCATE_3	0	0.00	0	0.00	4	0.88
TRUNCATE_2	0	0.00	0	0.00	4	0.88
TRUNCATE_1	59	2.85	72	3.23	96	5.13
BROKEN_CHAIN	1	0.41	1	0.41	58	2.83
SHUFFLED	14	1.53	12	1.43	86	3.80
MISLEADING	0	−4.36	0	−4.06	0	−5.13
FULL	96	5.13	96	5.13	96	5.13
Validity share vs IRRELEVANT	98.3%		98.6%		40.4%	
Validity share vs NONE	98.3%		98.6%		51.4%	

Direction separates the arms; distance does not. Accuracy does not decline with the gap: on the 4-bit checkpoint it is 18.1% at gap 1 and 24.0% at gap 7, though that cell is 6 of 25, so we draw no trend from it beyond the absence of visible harm (fig. 3). Reversing them does. Rule *after* its premise gives 26.8%, 22.4% and 94.9% across the three checkpoints against 5.8%, 4.8% and 84.9% for rule *before*; paired within item on the 4-bit checkpoint, 45 items favour rule-after against 2 ($p = 1.6 \times 10^{-11}$).

The direction of that effect holds on all three checkpoints, but its size does not: a 4.6–4.7 $\times$ ratio on the Qwen checkpoints is a 10pp modulation on Gemma-3-4B, which is near ceiling throughout. So ordering is not a *requirement* everywhere; what generalises is that reversing the pair costs accuracy, not that the model cannot proceed without it. Canonical order is better again (61.5% against $\sim 27\%$), so the chain before the last step matters too.

5.4 The validity share does not generalise

The decomposition of section 3.2 was run on two further checkpoints: the same model at bfloat16 rather than 4-bit, which isolates quantisation from model identity, and a different family. Table 3 carries all three; fig. 4 plots the split.

Quantisation is the smaller effect: bfloat16 raises TRUNCATE_1 from 59 to 72 of 96 while leaving the share essentially unchanged. One pair supports no general claim about precision. Model identity does not behave that way. Gemma-3-4B answers BROKEN_CHAIN correctly on **58 of 96** items where the other two manage 1, so the same control that isolates inference on one checkpoint exposes substantial surface exploitation on another.

Which baseline. The share divides by the effect against IRRELEVANT, which on the Qwen checkpoints also scores 0/96, so the choice is invisible there. Gemma-3-4B answers 22 of 96 entailed items with no certificate at all, so against NONE its share is 51.4%, not 40.4%. Both are in table 3, and we make no claim that most of its effect is surface overlap. **The control is also not inert on such a substrate:** Gemma scores 2/96 under IRRELEVANT and 4/96 under TRUNCATE_2 against 22/96 unaided, so an uninformative record costs it most of its own competence, and any effect measured against IRRELEVANT on a competent substrate is inflated by that cost. The divergence is the finding, not either number.

Order-sensitivity fails to replicate alongside the share: SHUFFLED scores 14 and 12 of 96 on the Qwen checkpoints and 86 on Gemma-3-4B. Reading a certificate as a bag of statements is a checkpoint-level property too.

We therefore do not claim that solver-supplied state is used as inference in general. On the checkpoint of section 3.2 it overwhelmingly is; on the strongest substrate in our sweep, between two-fifths and half survives. What generalises is the *method*, not the value it returns.

One result does replicate without exception, and it is the negative one (section 5.5).

5.5 Following a corrupted apparatus

The MISLEADING arm supplies a chain identical to FULL except in its last two sentences: a fabricated final rule, absent from the theory, and the terminal literal it licenses, both inverted. The ground-truth answer is unchanged. The model scores 3 of 192 against FULL’s 192, a paired difference of −98.4pp on all items and −100pp on the entailed

subset, where the exact test gives $p = 2.5 \times 10^{-29}$, with $d' = -4.363$. It replicates without exception: the arm scores 3, 6 and 0 of 192 across the three checkpoints, at d' of -4.36 , -4.06 and -5.13 , so the model exploiting surface overlap most is also the one most completely misled.

The class split is the sharper evidence. This model’s standing prior is No: under NONE it answers No to all 192 items, at $c = 2.565$. Under MISLEADING it answers Yes on **93 of the 96 contradicted items**, inverting a near-total prior because a fabricated rule told it to. Corrupted solver output does not merely fail to help; it overrides the model’s own strong default.

The model therefore follows a valid-looking certificate that points the wrong way on 189 of 192 items. Against the 98.3% share the two are jointly informative: what the model does with a supplied derivation is sensitive to whether it reaches the query, and offers no defence when it reaches the wrong conclusion. Section 5.6 shows that sensitivity is completion rather than verification, so the certificate is followed, not checked.

5.6 Abstention does not identify detection

A natural reading of section 3.2 is that the model detects a broken chain. The binary design cannot support that reading, because the correct answer under a broken certificate *is* the model’s default label: “detected the break” and “found no pattern to complete” predict the same response.

We separated them by rescoring the BROKEN_CHAIN items with three candidates rather than two. A validity tracker abstains *more* when the chain is broken; a pattern completer falls back to its default, which IRRELEVANT and NONE fix from either side. Table 6 and fig. 5 give all three checkpoints on all four arms.

The sign of the effect depends on the reference. Against the two records carrying no query-relevant material, NONE and IRRELEVANT, abstention falls on every checkpoint. Against TRUNCATE_1, the surface-matched *valid* partner of section 3.2, it *rises*, by 31.2 and 11.5pp on the two checkpoints with any abstention headroom. Neither comparison is clean. The unmatched references differ from BROKEN_CHAIN in content as well as validity, and are themselves 4.7pp apart at 4-bit against 34.9pp at bfloat16; TRUNCATE_1 is matched but is an arm the model can complete, so its low abstention reflects success, not a neutral baseline. **We therefore draw no conclusion from the abstention delta, and withdraw the claim that it refutes detection.**

What does not depend on a reference is what the strongest substrate does with a broken certificate: it abstains **0 times in 192** and answers Yes on 76. The 4-bit checkpoint shows the same behaviour in its counts: No is

110 of 192 under **both** BROKEN_CHAIN and TRUNCATE_1, the arms differing only in Yes, 1 against 61.

We find no positive evidence that any checkpoint verifies validity, but we no longer claim this test refutes detection. The direct evidence is section 5.5, where all three follow a fabricated final rule to the wrong answer, which a model checking validity would not. Section 6.1 notes the design limit behind this.

5.7 Which models can serve as the interface

Experiment 3 scores ten models on the three-class panel in all five arms; section 5.8 adds two more on a second runtime, twelve in all, nine common to both (one of the ten does not load there). We had declared that a model relays non-determination at $\geq 80\%$ recall on undetermined items with full solver material. **Eight of ten passed, and the criterion was invalid:** single-class recall is maximised by answering Unknown to everything, which four models approach: they emit Unknown on 54–97% of items averaged over arms, and never emit No at all. We replace it **post hoc** with a floor on every class, minimum per-class recall ≥ 0.50 , retaining both verdicts in the released results. A second criterion was prespecified and we report it here rather than let it disappear: a per-model paired FULL – NONE on the 64 undetermined items, exact McNemar, target 20pp. Five of ten meet it under Holm (+53.1, +48.4, +46.9, +37.5 and +20.3pp; the other five reach $\leq +9.4$ pp and none survives adjustment). Two of those five are rejected by the replacement criterion, at 0.375 and 0.422 minimum per-class recall: the same gameability by a different route. Table 4 gives all five arms for every model with its per-arm verdict.

Viability is a property of (model $\times$ arm $\times$ runtime), not of a model. The third factor is section 5.8’s. The decisive row is Qwen2.5-7B: minimum per-class recall falls from 0.938 under CONCLUSION_ONLY, where it is second best in the sweep, to 0.141 under FULL. Its contradicted-class recall collapses *when the proof state is supplied*: it can express three outcomes, and the state is what breaks it. **Supplying proof state degrades three of ten models** here, two of them otherwise among the strongest substrates. On the CUDA re-scoring the same measure gives two of eleven, which is section 5.8’s runtime factor again.

This bounds a claim we would otherwise have made. Nothing below 3B clears the floor in any arm, but that is a floor for *tolerating full certificates*, not for serving as an interface, and the effect is not monotonic. On the Qwen2.5 ladder to 14.7B, family and recipe fixed and the whole ladder on the second runtime, the harm runs 0.000, 0.000, -0.219 , -0.688 , $+0.062$: it peaks in a mid-range band and is *absent* at the top. Neither new model clears the floor, though. Llama-3.1-8B is **degenerate in every arm** de-

Table 4: All five arms, all twelve models. Each cell is balanced accuracy (% , chance = 33.3) / minimum per-class recall; bold clears the 0.50 recall floor, so VIABLE IN is derivable from the cells beside it. Shaded rows are the scale extension on a second runtime (section 5.8): the parameter ladder continues, the numbers are not comparable across the rule. Viability is per arm because it is a property of (model $\times$ arm $\times$ runtime), not of a model.

Model	B	NONE	IRREL	CONCL	PREFIX	FULL	VIABLE IN
Qwen2.5-0.5B	0.5	35.4/0.00	34.9/0.00	41.1/0.00	43.2/0.00	63.5/0.00	—
OLMo-2-1B	1.0	35.9/0.00	33.3/0.00	53.6/0.00	34.9/0.00	42.2/0.00	—
Llama-3.2-1B	1.2	33.9/0.00	33.9/0.00	57.8/0.00	36.5/0.00	37.0/0.00	—
Qwen2.5-1.5B	1.5	33.3/0.00	33.3/0.00	43.2/0.00	34.9/0.00	37.0/0.00	—
SmolLM2-1.7B	1.7	33.3/0.00	33.3/0.00	45.3/0.00	39.6/0.00	69.3/0.38	—
Qwen2.5-3B	3.0	41.1/0.00	39.1/0.00	96.4/0.91	74.0/0.56	90.6/0.72	concl, prefix, full
Llama-3.2-3B	3.2	45.8/0.00	43.2/0.02	67.2/0.02	55.2/0.02	80.7/0.42	—
Phi-4-mini	3.8	44.8/0.00	39.6/0.00	68.2/0.05	71.9/0.33	90.6/0.72	full
Gemma-3-4B	4.3	63.5/0.36	51.0/0.06	100.0/1.00	84.4/0.53	96.4/0.89	concl, prefix, full
Qwen2.5-7B	7.6	33.3/0.00	33.3/0.00	97.9/0.94	67.2/0.02	71.4/0.14	concl only
Llama-3.1-8B	8.0	38.5/0.00	34.4/0.00	65.6/0.00	64.6/0.00	66.7/0.00	—
Qwen2.5-14B	14.7	57.3/0.00	35.4/0.00	80.7/0.42	70.3/0.12	82.8/0.48	—

spite balanced accuracies up to 66.7%, the strongest single datum against a size floor. Qwen2.5-14B peaks at 0.484, so “absent at the top” is not “14B works”.

5.8 The runtime is a factor

Everything above is scored through `mlx-lm`. Re-scoring the same sealed panel, same weights and same prompt bytes through `transformers` on CUDA covers nine of the ten: GEMMA-3-4B’s weight mirror is an MLX conversion the second stack cannot load, which is itself a limit on how far a table can be moved between backends.

Eight of the nine agree on 95.5–99.7% of individual responses with no verdict moving. **Phi-4-mini does not.** It agrees on 87.9% (116 flips of 960), and its FULL arm is viable under one stack and not the other: minimum per-class recall 0.719 against 0.219, balanced accuracy 90.6% against 74.0%, on identical weights. The flips reach 3.0 logits, so this is not tie-breaking noise.

Two consequences. Viability is a property of (model $\times$ arm $\times$ runtime), not of a model. And the checkpoint carrying the 40.4–51.4% share of section 5.4 is GEMMA-3-4B, the one model this check cannot cover, so the share itself has no runtime replication, and we do not claim one.

5.9 The same 2×2 across ten models

Experiment 3’s five arms are the 2×2 of section 3.1 plus a baseline, on ten models, from receipts already collected. **This reading is post hoc**, like section 3.1’s. The mean state main effect is +8.7pp against a mean answer main effect of +21.6pp, with a mean interaction of −15.9pp; CONCLUSION_ONLY exceeds PROOF_PREFIX in 8 of 10 models and equals or exceeds FULL in 6 of 10. The answer chan-

nel dominates the state channel across the model set, and Experiment 1’s state main effect of +22.4pp is not typical of it. That mean averages over four responders degenerate in every arm, three with negative state effects; excluding them gives +14.7pp against +28.5pp, so the cross-model state effect is not less than half the single-model estimate and we do not claim it is. The ordering survives either subset.

6 Limitations

6.1 Scope of the mechanism claim

On both Qwen checkpoints the inference is *exactly one step deep*: TRUNCATE_2 scores identically to supplying no certificate (0/96), so the model completes a final inference when both premises are present and cannot chain two steps. Gemma-3-4B is not in that position (4/96 against 22/96 unaided), so depth is checkpoint-specific. Section 5.3 tests the ordering directly: reversing the final rule and its premise costs accuracy on every checkpoint, while separating them does not, which rules out the adjacency reading we first gave this. Nor does it *detect* invalidity, which section 5.6 tests directly and cannot establish, in either direction. Section 5.4 bounds the claim further: against the IR-RELEVANT control the share of the state effect that survives destroying validity spans 40.4–98.6% against IRRELEVANT, 51.4–98.6% against unaided.

The full theory is visible in every arm, so the query stays derivable whatever the certificate says, and breaking a certificate is diagnostic only where the model scores 0/96 from the theory alone, true of both Qwen checkpoints and not of Gemma-3-4B. A model using a broken certificate

as a pointer back into the theory would be doing inference of a different kind; on Gemma-3-4B that is a live alternative rather than a hypothetical one, since 20 of its 22 unaided successes recur under `BROKEN_CHAIN`. We record it as an interpretation caveat, not a controlled alternative. `TRUNCATE_2` and `TRUNCATE_3` also drop the query predicate, which lives in the final rule, so the depth ladder confounds depth with predicate presence and must be read against `SAME_ENTITY_IRRELEVANT`.

6.2 Post hoc analyses and superseded criteria

Four analyses are post hoc and labelled where used: the 2×2 reading of Experiment 1 is a different analysis of sealed data, not a re-run; Experiment 3’s viability criterion replaced a prespecified threshold the data showed to be invalid (the original is retained in the released results); and section 5.3’s gap-versus-direction split is read off eight orderings per item that happened to land in each cell, not a design that fixed them.

Experiment 1 also carries five defects we report rather than repair: the unreachable 15pp target noted above; no recorded proof depth, so its failures cannot be stratified; task identifiers encoding the semantic class; a release as a flat receipt export with no sealed panel, so its answer key is reconstructed from those identifiers rather than read from an authority; and a screen for response tokens leaking into a certificate that was a malformed regular expression and never fired (the released certificates contain none, checked after the fact). Experiments 2–4 fix all five.

6.3 Generalisation

One task family, small models, synthetic panels. Nonce vocabularies establish item novelty, not independence from every relevant pretraining pattern [Golchin and Surdeanu, 2023, Deng et al., 2024], and model is not randomised, so cross-model comparisons are descriptive.

Prompt format. Experiment 4 changes only the instruction and candidate set on byte-identical inputs, and the response distribution moves substantially [Zhao et al., 2021, Zheng et al., 2023]. Its arms share a baseline under that instruction, so the contrast holds but the absolute rates do not.

Quantisation. Experiments 1 and 2 use a 4-bit checkpoint and Experiment 3 bfloat16, so they describe different artefacts of the same model. Section 5.4 bounds the difference; we do not treat the two as one.

Prior report. Experiment 1’s per-arm counts and prespecified primary contrast appeared in an earlier unpublished report by the present authors; everything else is new. That report concluded the 3B checkpoint failed an open-

world unknown gate, whereas Experiment 3 finds the same family viable. Panels, candidate sets and quantisation all differ, which plausibly explains it, but we flag it rather than leave it to a reader who finds both.

An undetermined `PROOF_PREFIX` cannot contain the query predicate, so such items offer fewer surface cues and a surface-matching model looks worse on them for reasons unrelated to abstention.

Scoring regime. Every response is a forced choice among verified single-token candidates. No deployed solver-augmented pipeline works this way, so transfer of these results to a generative one is untested.

Checkpoint selection. Experiment 2’s three checkpoints were chosen *after* Experiment 3 identified viable substrates and cover two families; Phi-4-mini, the third viable substrate, was never run on those arms.

d' ceiling. At $n = 96$ the loglinear correction censors d' at ± 5.13 , and eight cells of table 3 sit on it, so d' *differences* involving a saturated arm, section 3.2’s 2.446 included, are bounds. Finally, every theory here arrives *already formalised*: nothing tests whether a model can turn a prose framework into solver input, which section 5.5 makes consequential.

7 Conclusion

Solver assistance improves a small model’s accuracy on rule-chaining queries, and end-to-end accuracy cannot say why. The decomposition can: arrange the arms so each contrast moves one factor, and break validity while holding surface form fixed. What it returns is model-specific: 98.3% of the state effect survives on one checkpoint and 40.4–51.4% on another, so we offer the method as the contribution and decline to generalise its value, ours included.

Three findings hold on every checkpoint, all negative. The inference is one step deep on the Qwen checkpoints, and degrades everywhere when the final rule precedes its premise. Detection we cannot establish either way: the abstention delta changes sign with the reference arm. And nothing defends against a wrong apparatus: the three answer incorrectly on 189, 186 and 192 of 192, inheriting its errors. Two criteria we had prespecified failed here: a recall floor on one class, gamed by models answering that class almost everywhere, and a 15pp target an arm at 91.7% could not reach. Both were caught only by reporting sensitivity and headroom beside accuracy, as was a baseline arm at 50.0% with $d' = 0.00$. We recommend that practice as standard.

Reproducibility Statement

Code and data. The supplementary material accompanying this submission is self-contained: sealed panels and answer authorities, all 34,584 per-response receipts, the generators, runners, audits and analysis scripts, and the forward-closure certifier with its validation against the public corpus. Every number in this paper can be re-derived from a fresh unpack with no network access, no model weights and no external packages; the audits use the Python standard library only, so verification does not depend on resolving an environment. Panels are hash-stamped, and the runners for Experiments 2 and 3 refuse a panel whose hash does not match. Each receipt carries a SHA-256 of its own contents, and all 34,584 verified with zero failures. The count is not typed into the manuscript: it is produced by `scripts/census_receipts_v1.py` from the released receipts and substituted at compose time.

Checking one number takes a minute. `scripts/verify.py` recomputes each headline value and prints it beside the value this paper states. Five of its seven claims re-derive from the receipts and the sealed authority; the cross-model mean and the response census read released analysis files, which the listed scripts regenerate: `python3 scripts/verify.py validity-share` recomputes the 98.3% of section 3.2 and shows the four counts it came from. `--all` does the same for the validity share, Gemma’s two baselines, the corruption counts, the detection direction on three checkpoints, the 2×2 state main effect, the cross-model mean and the response census. It uses the standard library only, so a reviewer needs no environment beyond Python. Model weights are not redistributed, but each run records the pinned revision and a hash of the weight files before those files are deleted. A permanent archive with a DOI will accompany the camera-ready version.

AI-Use Statement

A large language model was used as a coding and drafting assistant throughout: generating experiment code, analysis scripts and figure code, and drafting this manuscript. All experimental design decisions, the interpretation of results, and the claim boundaries were made and verified by the authors. Every numeric value reported was produced by the released analysis scripts from sealed receipts, not written by a model; the released audits allow any reader to re-derive them independently.

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A Supporting Tables and Figures

Supporting material for the results in the body. Table 5 inventories the sealed panels; table 1 gives the four edges of the 2×2 with their adjusted p values; table 6 the abstention deltas of section 5.6. The four figures plot results whose values are already tabulated above: the ten arms (fig. 2), the validity share across checkpoints (fig. 4), the three-class response distribution (fig. 5), the order profile (fig. 3) and the substrate ladder (fig. 6).

Table 5: Datasets. The sealed panels are nonce-generated and disjoint; the public corpus validates the certifier only. Experiment 4 rescores Experiment 2’s items rather than generating new ones.

Panel	Classes	Items	Used by
b14	entailed / contradicted	192	Experiment 1
b15	entailed / contradicted	192	Experiment 2
b16	entailed / contradicted / undetermined	192	Experiment 3
b17, b17b	b15 items, three candidates	192	Experiment 4
b18	b15 items, 8 line orders each	192	section 5.3
b16 on a second stack	as b16	192	section 5.8
ProofWriter OWA [Tafford et al., 2020]	three-way	23,240	certifier validation

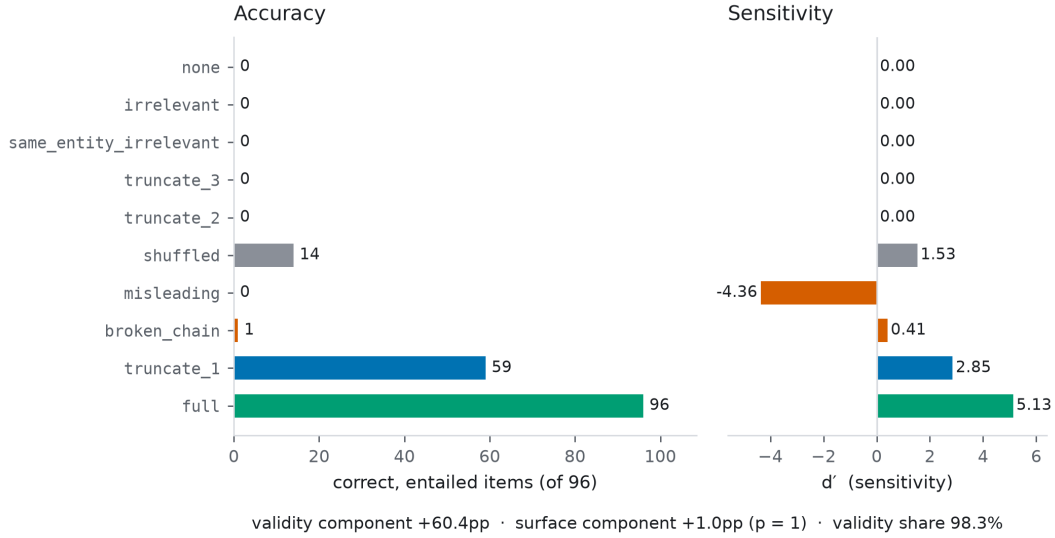


Figure 2: Ten matched arms on one item set, differing only in the solver record. Left: correct on the entailed subset (of 96). Right: sensitivity d' , signed. Destroying validity alone removes almost the whole effect *on this checkpoint*; on gemma-3-4b it does not, see section 5.4. The MISLEADING arm, whose fabricated final rule points the wrong way, is followed to the wrong answer on 189 of 192 items.

Table 6: Three-candidate rescoring: the change in **Unknown** rate under **BROKEN_CHAIN**, in points of 192, against each of three references. Detection predicts a *rise*. It falls against the two records carrying no query-relevant material and rises against the surface-matched **TRUNC_1**, so the sign is a function of the reference. Per-arm rates are in section 5.6.

Checkpoint	Δ_{NONE}	$\Delta_{\text{IRREL.}}$	Δ_{TRUNC_1}
Qwen2.5-3B 4-bit	-37.5	-32.8	+31.2
Qwen2.5-3B bf16	-3.6	-38.5	+11.5
Gemma-3-4B	-0.5	-7.8	0.0

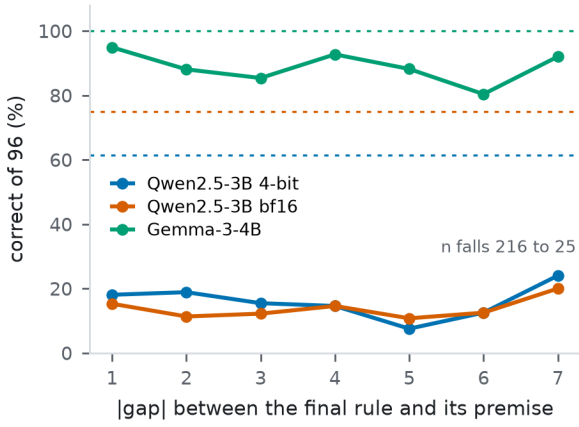
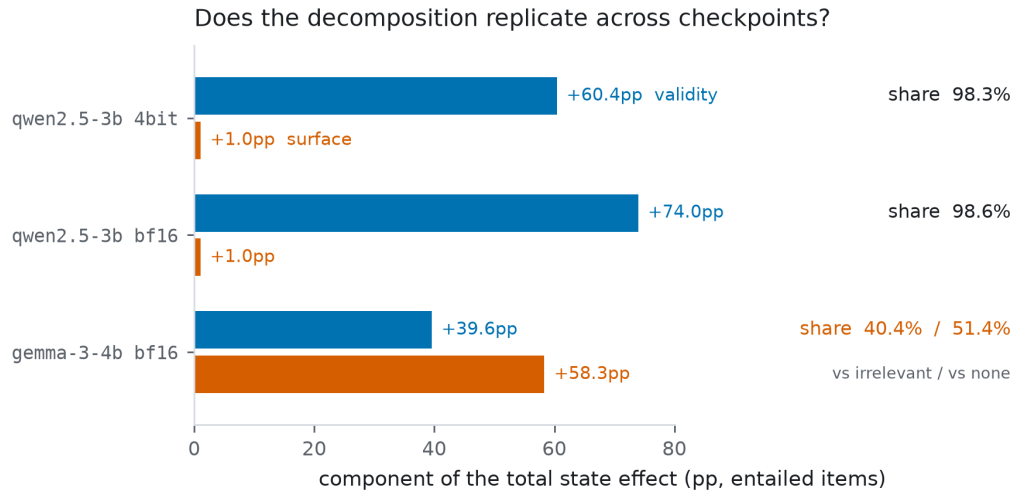
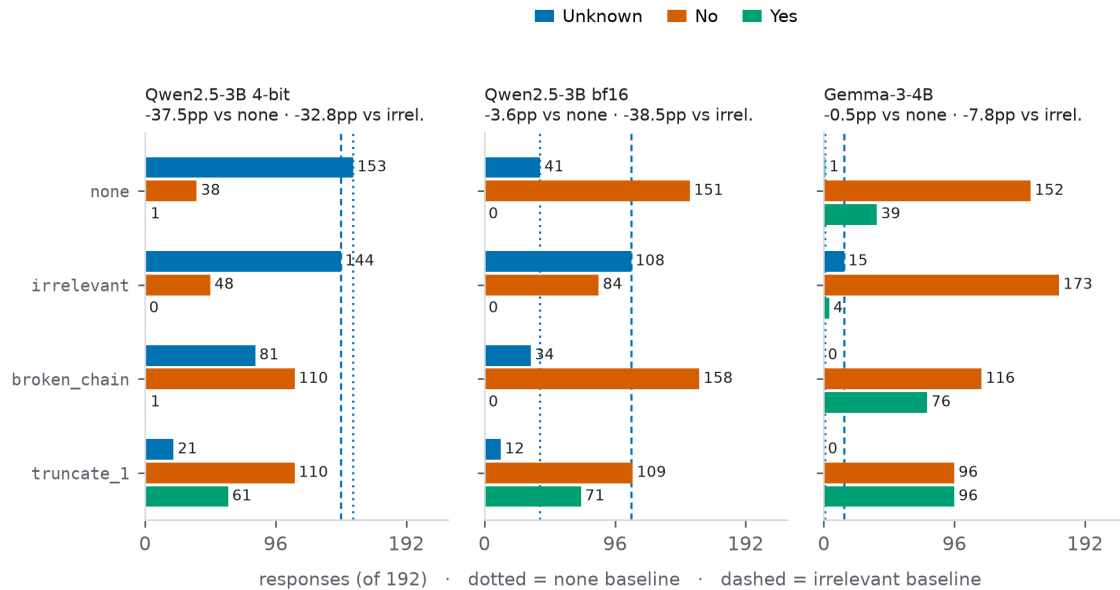


Figure 3: Accuracy against the gap between the final rule and the premise it fires on: all seven gaps, all three checkpoints, each checkpoint’s canonical-order score dashed. Distance is flat. Direction is not: rule-after scores 26.8, 22.4 and 94.9% against 5.8, 4.8 and 84.9% for rule-before. Gemma-3-4B still answers 84.9% of rule-before items correctly, so reversal degrades rather than prevents.



The 98.3% share does not generalise: gemma-3-4b's surface component is +58.3pp against +1.0pp on both qwen checkpoints. Its share is 40.4% against the same-shape control and 51.4% against what it manages unaided — the two denominators diverge only where a substrate has competence of its own.

Figure 4: The surface and validity components on three checkpoints. Gemma-3-4B's surface component is +58.3pp against +1.0pp for the two Qwen checkpoints. Bars give the decomposition against the IRRELEVANT control; the share annotation gives both denominators, 40.4% against that control and 51.4% against what the model manages unaided. The two coincide on the Qwen checkpoints, which score 0/96 with no certificate, and diverge only where a substrate has competence of its own.



On every checkpoint, against either reference, a broken chain lowers abstention rather than raising it. Gemma-3-4B abstains on 0 of 192 broken chains and completes 76 of them.

Figure 5: Three-candidate response distribution, three checkpoints, four arms. Dotted and dashed lines mark the NONE and IRRELEVANT abstention baselines, 4.7pp apart at 4-bit and 34.9pp at bfloat16. A broken chain lowers abstention against *both* references on every checkpoint, the opposite of the direction validity detection predicts.



Figure 6: All five arms for all ten models, ordered by parameter count. Chance is 33.3%. Bar colour gives the per-arm viability verdict. Qwen2.5-7B is viable under CONCLUSION_ONLY and not under FULL: viability is a property of (model $\times$ arm $\times$ runtime), not of a model.